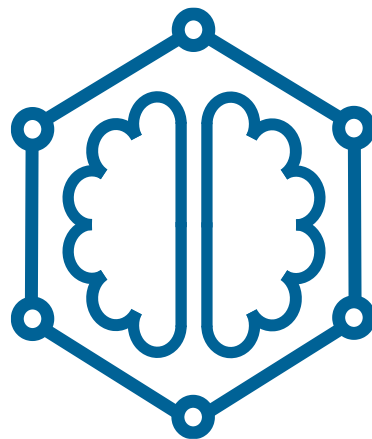




What is a Knowledge Graph?

Choosing a Definition

Emanuel Sallinger



“A graph that understands real-world entities and their relationships to one another:
things, not strings”

““Introduction the Knowledge Graph: Things, Not Strings”. A. Singhal. Google Blog 2012.

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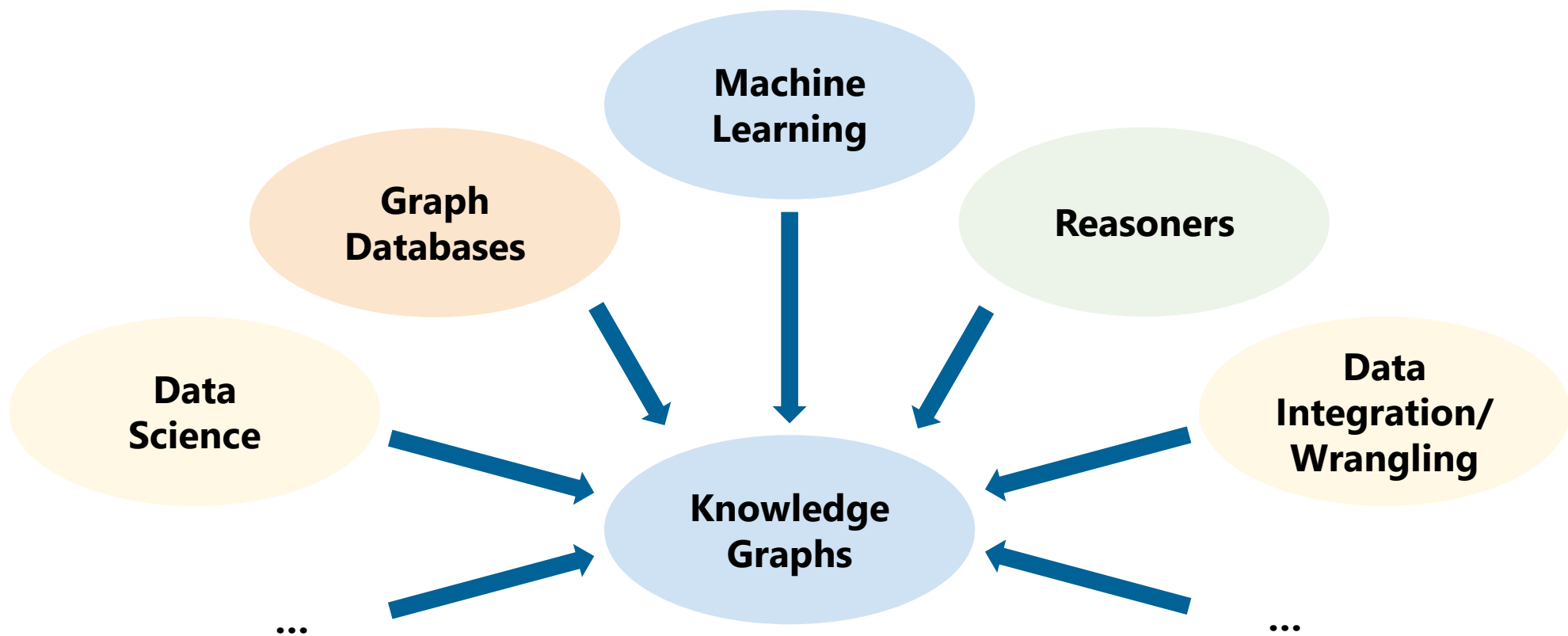
“**A knowledge graph** acquires and integrates information into an **ontology** and applies a **reasoner** to derive new knowledge”

“Towards a Definition of Knowledge Graphs”. L. Ehrlinger and W. Wöß. SEMANTICS 2016.

result of the application of the inference rules over the ground extensional component (with the so-called “reasoning” process)”

n et al. Dagstuhl 2019.

“Knowledge Graphs and Enterprise AI: the Promise of an Enabling Technology”. L. Bellomarini et al. ICDE 2019.





Facebook KG Social graph with people, places, things



Amazon PG knowledge graph of all products



Factual Businesses & places



Wolfram KB World facts + mathematics



RIT People, skills, recruiting



Commercial Banks Customers, Companies, Risks, ...



Rating Agencies Companies, Evaluations, Risks, ...

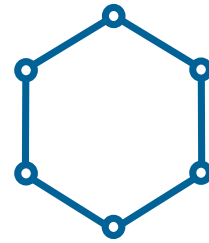


Starting at the Core

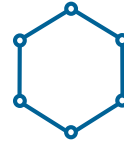




Knowledge



Graph



Graph



Knowledge

?

$$KG = (G, K)$$

...

Relational Data

Property Graphs

XML

RDF*

JSON

CSV Files

RDF



Knowledge



$$KG = (D, K)$$



$$KG = (D, K; g)$$

$$g: \mathbb{D} \rightarrow \mathbb{G}$$

$$KG = (\overline{D}, \overline{K}; \overline{g})$$

$$g: \mathbb{D} \rightarrow \mathbb{G}$$

$$KG = (\overline{D}, \overline{K}; g, \dots)$$

$$g: \mathbb{D} \rightarrow \mathbb{G}$$

$$KG = (D, K, \{g, \dots\})$$

$$g: \mathbb{D} \rightarrow \mathbb{G}$$

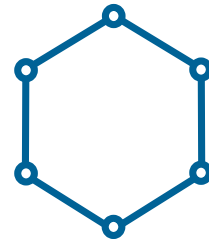


$$KG = (D, K; g)$$

$$g: \mathbb{D} \rightarrow \mathbb{G}$$



Knowledge



Graph

... Entity Disambiguation

Link Prediction Query Answering

Entity Classification

Question Answering ...

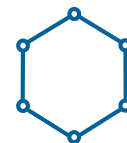


Making Knowledge Work



Obtaining Knowledge

(Learning* Knowledge)



Knowledge Graph

$$KG = (D, K; g, l)$$

Obtaining
Learning

Obtaining Knowledge

(Learning* Knowledge)



Knowledge



Graph



$$KG = (D, K; g, l)$$

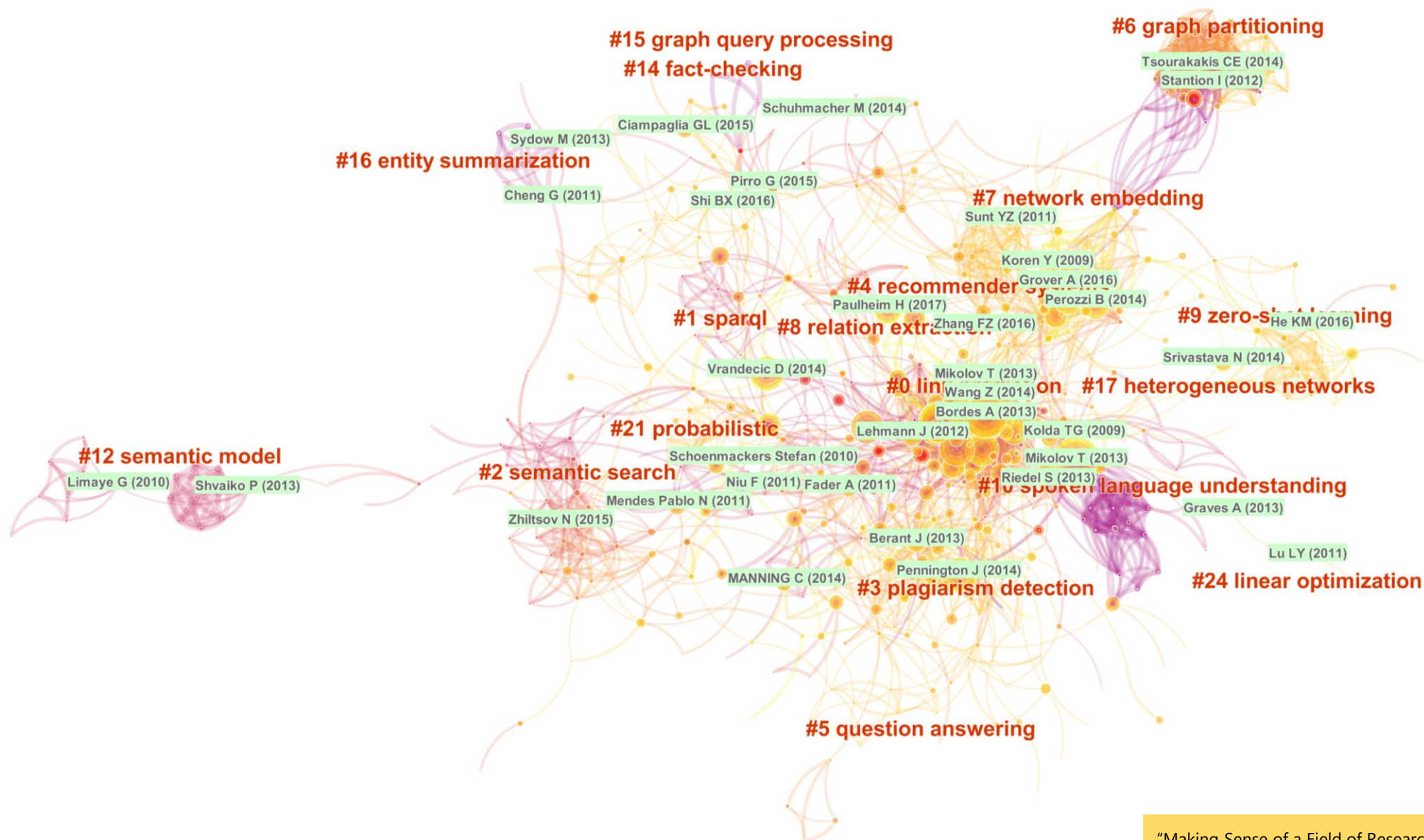
$$l_{\mathbb{K}}: \mathbb{D} \rightarrow \mathbb{K}$$

... Entity Disambiguation

Link Prediction Query Answering

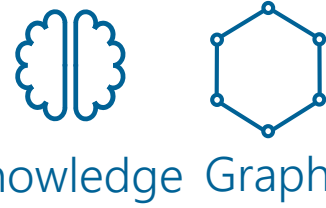
Entity Classification

Question Answering ...



Applying Knowledge

(Reasoning* with Knowledge)



$$KG = (D, K; g, l, r)$$

Obtaining
Learning

Applying
Reasoning

Applying Knowledge

(Reasoning* with Knowledge)



Knowledge



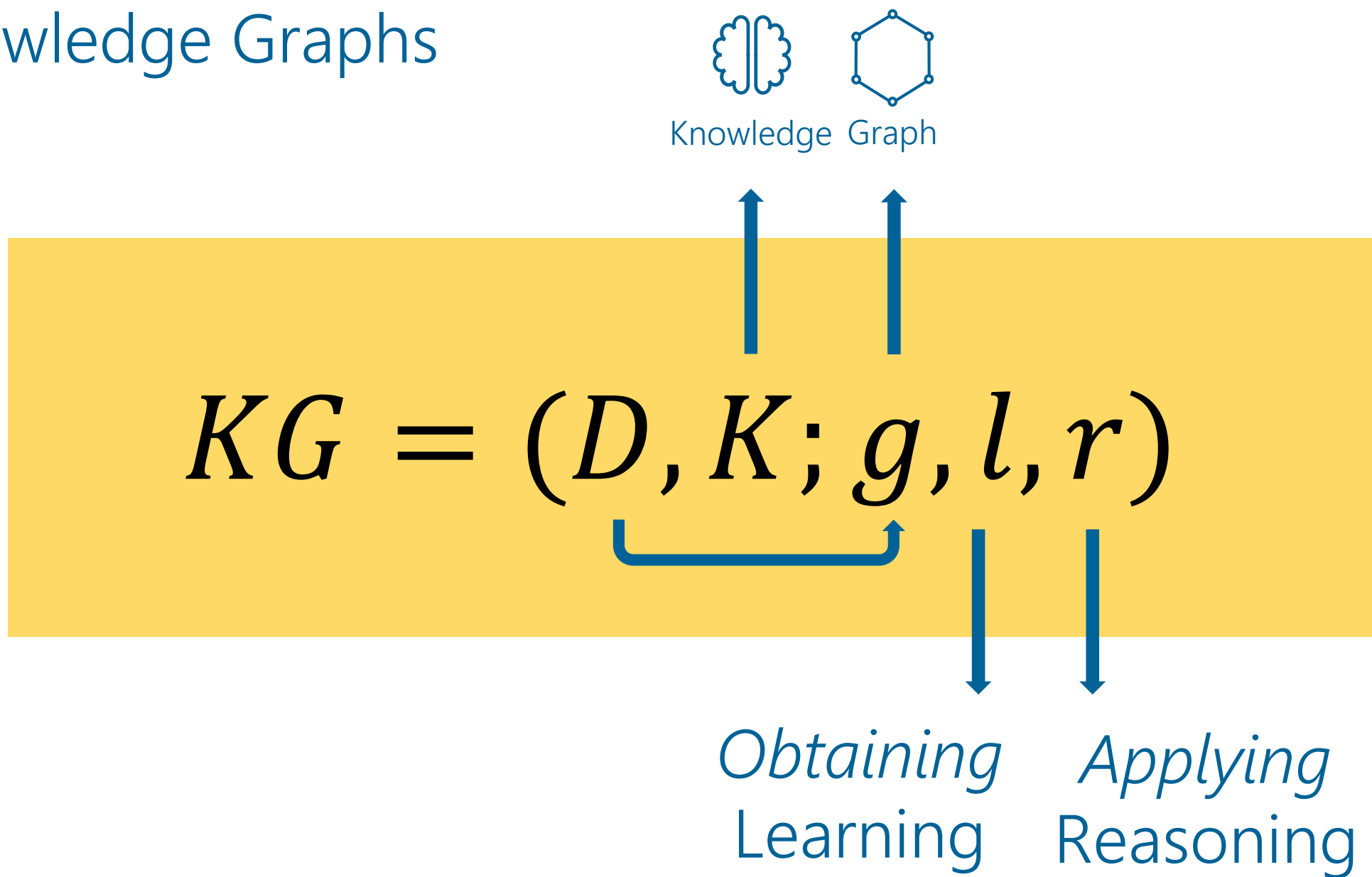
Graph



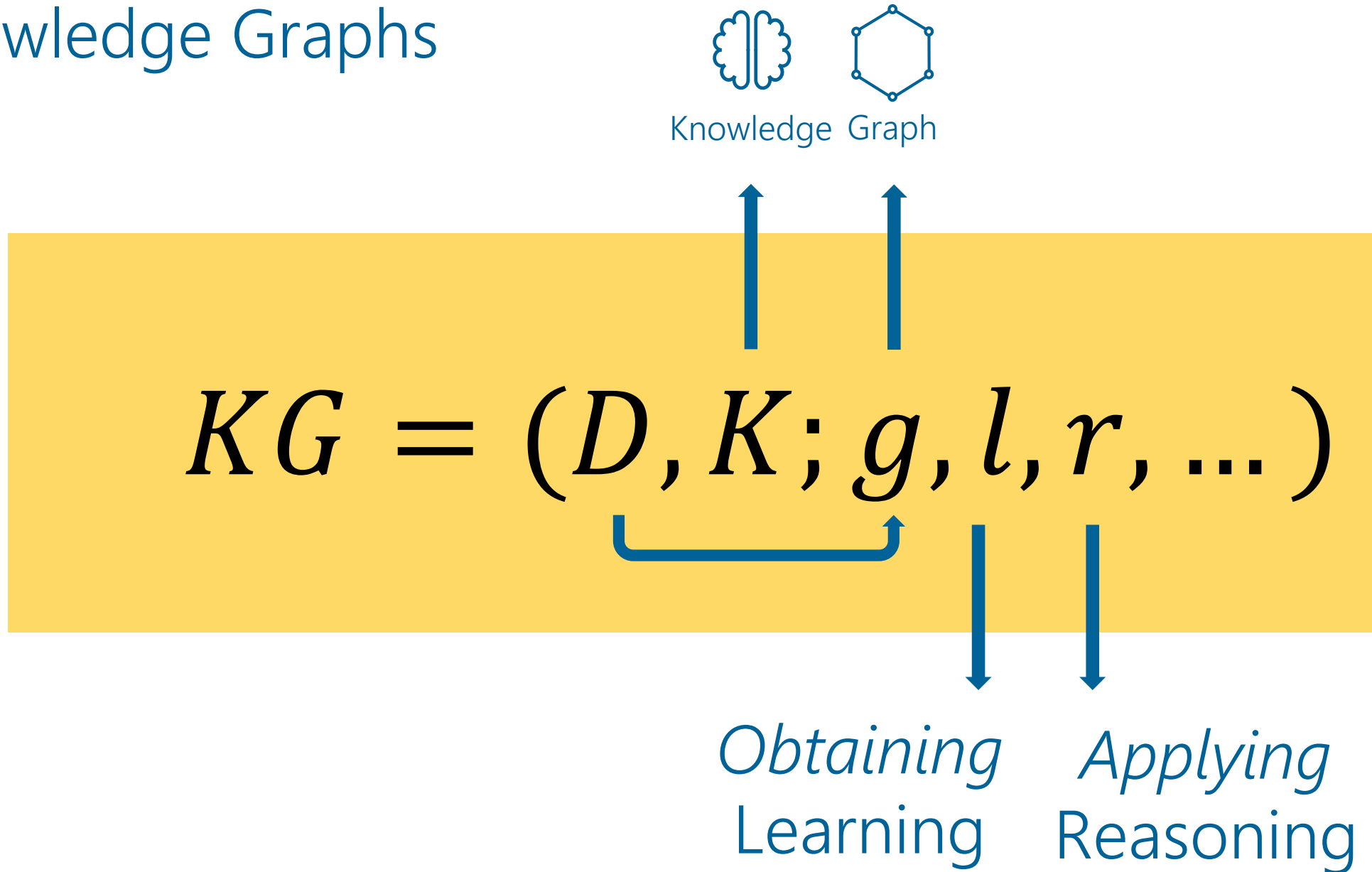
$$KG = (D, K; g, l, r)$$

$$r_{\mathbb{K}}: \mathbb{D} \rightarrow \mathbb{D}$$

Knowledge Graphs



Knowledge Graphs





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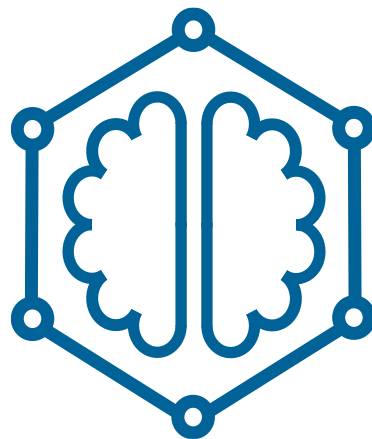
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